Ethan Bobby Kurien

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.83
Indian Education School, Kuwait	Class XII (CBSE)	2023	98.60%
Indian Education School, Kuwait	Class X (CBSE)	2021	99.40%

SCHOLASTIC ACHIEVEMENTS

- **Bank of Baroda Excellence in Academics Award:** Highest CGPA in Batch '27 *(FY 2024-25)*
- **Certificate of Academic Excellence, IITH** *(2023-24, 2025-26)*
- **Selected for Double Major Program in Computer Science** (Double Major CGPA of 9.5)
- **Scored 8.5/9 in IELTS and 1540/1600 in SAT (800/800 Math)**, placing in the top 1% globally.
- **Grade 10 & 12 CBSE Board School Topper** *(2021 & 2023)*; **Kuwait Gold Medalist**, Indian Business Council

EXPERIENCE

Applied AI Research Intern, Thomson Reuters, Bengaluru (Jan - Jul 2026)

- Built an agentic retrieval pipeline (LangGraph orchestration, Langfuse-tracked evaluation) with a multi-pass LLM-as-judge, improving ranking precision 3x over the deployed baseline.
- Built a knowledge-graph-augmented retriever matching frontier-model accuracy at ~25% of inference cost via a two-tier gate on AWS Bedrock, plus a fine-tuned 4B open-weight model (QLoRA/Unsloth) as a relevance classifier.
- Built a ground-truth benchmark from scratch using hybrid keyword/semantic matching and a web-search sourcing agent, 6x larger than the existing hand-curated benchmark.
- Led a 135K-call study (tracked with Braintrust) across 9 models and 15 tasks, showing that chain-of-thought reasoning reduces answer reproducibility on subjective tasks; paper submitted to KSE Conference, 2026.
- Skills: Python, LangGraph, AWS Bedrock, Langfuse, Braintrust, Unsloth, Agentic RAG, Knowledge graphs

Digital Innovations Intern, Kamco Invest, Kuwait City

- Built a RAG chatbot for internal knowledge access by scraping the Kamco Invest website (requests, BeautifulSoup, ChromaDB), and a time series Bitcoin price-prediction model (H2O AutoML) generating Buy/Sell/Hold signals.

PROJECTS

Physically Informed MissForest (PIMF) for Streamflow Gap-Filling | ML Research (Python, Gradio, HF)

Supervised by Dr. Maheshwaran, Climate Change Department; Paper submitted to *Applied Soft Computing Journal*

Built the PIMF framework using a Weighted Random Forest integrating hydrological connectivity and spatial-temporal features to reconstruct missing streamflow data. Achieved 15-20% accuracy improvement (NSE: 0.80-0.91) over baselines across 3-100 day gaps. Deployed an interactive Gradio app on Hugging Face Spaces for automated streamflow gap-filling.

Evenly | Full-Stack Web App (Next.js, TypeScript, Supabase, PostgreSQL)

Built a full-stack expense-splitting app with in-browser OCR (Tesseract.js) and an LLM layer (Llama 3.3) for receipt/text parsing, plus a greedy debt-simplification algorithm cutting group settlements to n1 payments.

cc-mini | AI Agent Systems (Python, LangGraph, Ollama)

Built a miniature coding agent with nine tool-calling functions, an OS-level sandbox (Seatbelt/bubblewrap), and a permission gate for file/shell/network actions; supports local open-weight models by default with Anthropic/OpenAI as optional providers.

Skyway | **Flight Route Planning** (C++, JavaScript, Dijkstra's Algorithm)

Implemented Dijkstra's algorithm over a 3,200-airport graph to compute shortest-distance flight routes with an optional max-hop constraint, shipping both a browser app with 3D globe visualization and a C++ CLI for terminal-based queries.

SKILLS

- **Languages:** Python, C++, C, SQL
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, Deep Learning (CNN, RNN, ViT), LLM Agents & Retrieval (RAG, LangGraph, ChromaDB), Fine-tuning & Evaluation, Random Forest, NumPy, Pandas, Matplotlib, Statistical Modeling
- **Systems & Backend:** Git, Next.js, Supabase, Flask, REST APIs, Streamlit, Gradio, MATLAB
- **Coursework & Certifications:** Data Structures & Algorithms, Operating Systems, Computer Architecture, Database Management Systems, Probability & Statistics, Linear Optimisation; IBM Data Science Professional Certificate (Coursera, 2025), Build AI Agents with LangGraph, MATLAB Fundamentals (MathWorks, 2024)

POSITIONS OF RESPONSIBILITY

- **Quadruped Robot Team, Overall Lead:** Leading the development of an obstacle-detecting autonomous quadruped robot; represented IIT-H at the prestigious Pan IIT Tech Expo, IIT Bombay.
- **Office of Career Services, Internship Coordinator:** Coordinating internship placements by engaging with companies, managing outreach, and streamlining student applications.
- **International Relations Cell, Outbound Programs Core:** Facilitated student participation in global academic and research exchange programs.

EXTRACURRICULAR

- **Sakura Science Program, University of Tokyo:** Represented IIT Hyderabad at a JST-hosted winter program at the University of Tokyo, featuring talks and lab tours on differential privacy algorithms and spintronics for AI chips.
- **UN Global Social Leaders Contest:** Ran an SDG-focused recycling drive, connecting 30+ homes to local recycling centers via a reward-based leaderboard built on Squarespace.
- **Media Assistant & Keyboard Player:** Band member at KTMCC, Kuwait City.